



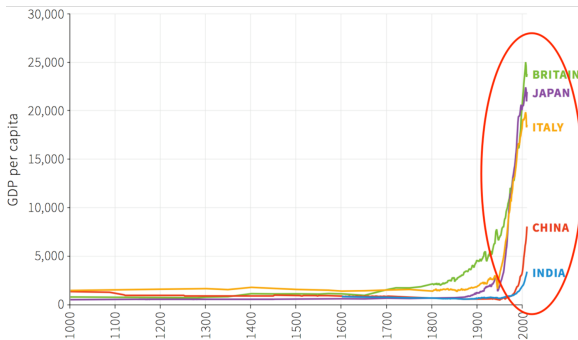
Advanced Macroeconomics

Section 2 - Growth (I): The mechanics of capital accumulation and growth

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The hockey stick of history



Section 2: Growth (I)

The Plan

1. Harrod-Domar
2. Solow
3. Ramsey-Cass-Koopmans
4. Diamond's overlapping-generations (OLG)

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The Plan

1. Harrod-Domar
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4. Diamond's overlapping-generations (OLG)
 - ▶ Models of the *mechanics* of growth
 - ▶ Interaction of capital accumulation, technological progress & production in a growing economy.
 - ▶ They can't really explain the hockey stick pattern or the deep fundamental causes of growth.

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- ▶ $\frac{\dot{X}(t)}{X(t)} = \text{growth rate of } X$
- ▶ g_X is a shorthand for $\frac{\dot{X}(t)}{X(t)}$

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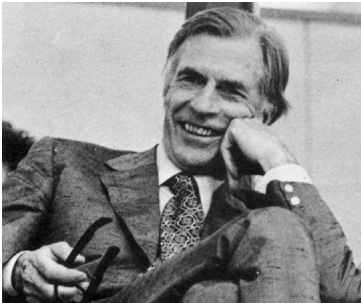
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 - *Dynamic stability* if over time any deviation from equilibrium tends to (quickly or slowly) disappear.

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- o 'Grandfather' of modern growth theory.
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- o **Premise 2**: investment depends on aggregate demand
- ▶ **Main findings**:
 - unique equilibrium path: $g_w = sa$ (*warranted rate*)
 - warranted rate does not guarantee full (nor stable) employment
 - instability: economy won't converge to g_w , except by a fluke

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- Rate of capacity utilization: $u(t) = \frac{Y(t)}{Y^*(t)}$
- Investment rate: $\dot{g}_K(t) = \alpha(u(t) - 1)$ with $\alpha > 0$

Harrod-Domar: Intertemporal equilibrium

Assumptions:

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Intertemporal equilibrium ('warranted' growth rate):

$$\dot{g}_K(t) = 0 \rightarrow u = 1 \rightarrow g_W = sa$$

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- ▶ More formally (by plugging $I=S$ condition into investment function):

$$\dot{g}_K = \alpha \left[\frac{g_K}{g_W} - 1 \right]$$

→ When the growth rate is above equilibrium ($g_K/g_W > 1$), it will tend to increase even more (and vice versa)

$$\dot{g}_K = \alpha \left[\frac{g_K}{g_W} - 1 \right] \quad \text{with } \alpha > 0$$

